

The Stereographic Codex: Complete Theorem Catalog

Every Machine-Verified Theorem in the Pythagorean Cosmos

The Stereographic Codex Research Series

2025

Table of Contents

Part 1: Theorem Catalogs

Part 1

Chapter 1.1

Source: FINAL_CATALOG.md

Complete Catalog of Machine-Verified Mathematics

Grand Unified Edition ? Final

"The equation $a^2 + b^2 = c^2$ is not one theorem. It is a portal."

How to Read This Catalog

Every theorem listed here compiles in Lean 4 with Mathlib. The checkmark (?) is implicit ? if it appears in this catalog, the computer verified it. The single exception is the Sauer?Shelah lemma, which is explicitly marked. Lean names in backticks (e.g., stereo_on_circle) are the exact identifiers in the source code. File names refer to .lean files in the project root.

Overview

Metric Value	-----
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The Central Thesis

Stereographic projection ? the map

$$[\sigma(t) = \left(\frac{1-t^2}{1+t^2}, \frac{2t}{1+t^2}\right)]$$

? is a canonical isomorphism linking six pillars of mathematics through the single identity $a^2 + b^2 = c^2$:

| Pillar | Reading of $a^2 + b^2 = c^2$ |

|-----

These are not analogies. They are the same mathematical object in different coordinate systems. Every claim is machine-checked.

Architecture

? : ? ? $S^1 \setminus \{?1\}$

(stere

The Seven Research Teams

| Team | Codename | Domain | Key Files |

|-----

Pillar I: The Universal Decoder

The single formula that connects everything.

Foundation Theorems

#	Lean Name	Statement	File
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---|

Decoder Channels (59 theorems in UniversalDecoder.lean)

#	Lean Name	Statement	File
---	-----------	-----------	------

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Plus 47 additional decoder channel theorems.

N-Dimensional Generalization

#	Lean Name	Statement	File
---	-----------	-----------	------

---|

Pillar II: The Berggren Tree

The infinite ternary tree generating every primitive Pythagorean triple exactly once.

Tree Structure

#	Lean Name	Statement	File
---	-----------	-----------	------

---|

Descent Theory

| # | Lean Name | Statement | File |

| --- | -

Landscape Theory

| # | Lean Name | Statement | File |

| --- | -

Pillar III: The Light Cone

Pythagorean triples ARE photon momenta.

Minkowski Fundamentals (42 theorems in LightConeTheory.lean)

| # | Lean Name | Statement | File |

| --- | -

Photonic Frontier (53 theorems in PhotonicFrontier.lean)

| # | Lean Name | Statement | File |

| --- | -

Pillar IV: The Crystal

Weights that crystallize onto the integer lattice.

Crystallization Dynamics

#	Lean Name	Statement	File
---	-----------	-----------	------

|---|-

Stability & Safety

#	Lean Name	Statement	File
---	-----------	-----------	------

|---|-

Pillar V: The Hurwitz Tower

Division algebras in dimensions $1 \leq 2 \leq 4 \leq 8$, and only those.

Composition Identities

#	Lean Name	Statement	File
---	-----------	-----------	------

|---|-

Hopf Fibration

| # | Lean Name | Statement | File |

| --- | -

Pillar VI: The Quantum Bridge

From Pythagorean rationals to universal quantum gates.

Gate Algebra

| # | Lean Name | Statement | File |

| --- | -

Pythagorean Gates

| # | Lean Name | Statement | File |

| --- | -

Cross-Domain Isomorphism Dictionary

| Domain A | Bridge Theorem | Domain B |

| -----

The Grand Unification Chain

$\mathbb{R}^2$ $\xrightarrow{\text{Euclid}}$ Pythagorean Triples $\xrightarrow{Q=0}$ Light Cone

?

Every arrow is a machine-verified theorem.

Inside-Out Factoring (IOF)

| # | Lean Name | Statement | File |

|---|

Application Summary

| Application | Core Theorem | Status |

|-----

The remaining ~2,500 theorems span 40+ domains of pure and applied mathematics.

Number Theory

NumberTheory · NumberTheoryAdvanced · NumberTheoryDeep · FLT4 · FermatFactor · CongruentNumber · DiophantineApproximation · AlgebraicNumberTheory

Algebra

Algebra · AlgebraicStructures · GaloisTheory · LieAlgebras · RepresentationTheory · RepTheoryDeep · CommutativeAlgebra · GeometricAlgebra · SL2Theory

Analysis

Analysis · AnalysisInequalities · FunctionalAnalysis · HarmonicAnalysis · DifferentialEquations · MeasureTheory · SpectralTheory · OperatorAlgebras · NumericalAnalysis

Topology & Geometry

Topology · AlgebraicTopology · DifferentialGeometry · SymplecticGeometry · MetricGeometry · ConvexGeometry · KnotTheory · HodgeTheory

Combinatorics & Graph Theory

Combinatorics (1 sorry: Sauer?Shelah) · ExtremalGraphTheory · SpectralGraphTheory · RamseyTheory · AdditiveCombinatorics · ArithmeticCombinatorics · MatroidTheory · CodingTheory

Category Theory & Homological Algebra

CategoryTheory · CategoryTheoryDeep · HomologicalAlgebra · AlgebraicKTheory

Logic & Foundations

SetTheory · SetTheoryLogic · ModelTheory · ComputabilityTheory · DescriptiveSetTheory · Complexity

Applied Mathematics

Probability · StochasticProcesses · InformationGeometry · Entropy · OptimizationTheory · GameTheory · MathBiology · CryptographyFoundations · DriftFreeIMU · HomingMissile

File	Domain	Theorem Count (approx.)
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* 159 Lean 4 source files

The Harmonic Number Theory Group

Chapter 1.2

Source: GRAND_UNIFIED_CATALOG.md

Complete Catalog of 2,637 Machine-Verified Theorems

Version 2.0 ? Grand Unified Edition

Every theorem in this catalog compiles in Lean 4 with Mathlib using only the standard axioms (propext, Classical.choice, Quot.sound). The single exception ? the Sauer?Shelah lemma ? is explicitly marked.

Project at a Glance

Metric	Value
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The Seven Teams

Team	Codename	Domain	Key Files
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I. THE DECODER ? Stereographic Projection & Universal Channels

The single formula $t \mapsto \bigl(\frac{1-t^2}{1+t^2}, \frac{2t}{1+t^2}\bigr)$ that connects everything.

A. Foundation Theorems

| # | Theorem | Statement | File |

| --- | -

B. Universal Decoder Channels

| # | Theorem | Statement | File |

| --- | -

+ 47 additional decoder channel theorems in UniversalDecoder.lean (59 total).

C. N-Dimensional Generalization

| # | Theorem | Statement | File |

| --- | -

II. THE TREE ? Berggren Structure & Pythagorean Triples

The infinite ternary tree generating every primitive Pythagorean triple exactly once.

A. Tree Structure

| # | Theorem | Statement | File |

| --- | -

B. Descent Theory

| # | Theorem | Statement | File |

| --- | -

C. Landscape Theory

| # | Theorem | Statement | File |

| --- | -

III. THE CRYSTAL ? Neural Architecture & Intelligence Crystallizer

Weights that crystallize onto the integer lattice.

A. Crystallization Dynamics

| # | Theorem | Statement | File |

| --- | -

B. Stability & Safety

| # | Theorem | Statement | File |

| --- | -

IV. THE LIGHT CONE ? Minkowski Geometry & Photon Physics

Pythagorean triples ARE photon momenta.

A. Minkowski Fundamentals (42 theorems in LightConeTheory.lean)

| # | Theorem | Statement | File |

| --- | -

B. Photonic Frontier (53 theorems in PhotonicFrontier.lean)

| # | Theorem | Statement | File |

| --- | -

V. THE TOWER ? Gaussian Integers, Quaternions & Hurwitz Hierarchy

Division algebras in dimensions 1 ? 2 ? 4 ? 8, and only those.

A. Gaussian Integers & Two-Square Identity

| # | Theorem | Statement | File |

| --- | -

B. Quaternions & Four-Square Identity

| # | Theorem | Statement | File |

| --- | -

C. Octonions & The Hurwitz Theorem

| # | Theorem | Statement | File |

| --- | -

VI. THE FACTORING ENGINE ? Inside-Out Factoring & Landscapes

The Berggren tree as an integer-factoring algorithm.

A. IOF Core

| # | Theorem | Statement | File |

| --- | -

B. Energy Descent

| # | Theorem | Statement | File |

| --- | -

VII. THE QUANTUM BRIDGE ? Gates, Bloch Sphere & Computation

From Pythagorean rationals to universal quantum gates.

A. Gate Algebra

| # | Theorem | Statement | File |

|---|-

B. Pythagorean Gates

| # | Theorem | Statement | File |

|---|-

VIII. THE MATHEMATICAL COSMOS ? 40+ Domains of Pure & Applied Mathematics

The remaining ~2,500 theorems span the following domains. Each file is a self-contained verified module.

Number Theory

| File | Highlights |

|-----

Algebra

| File | Highlights |

|-----

Analysis

| File | Highlights |

|-----

Topology & Geometry

| File | Highlights |

|-----

Combinatorics & Graph Theory

| File | Highlights |

|-----

Category Theory & Homological Algebra

| File | Highlights |

|-----

Logic, Set Theory & Foundations

| File | Highlights |

|-----

Applied Mathematics

| File | Highlights |

|-----

IX. THE ROSETTA STONE ? Cross-Domain Bridge Theorems

The theorems that reveal the deep unity.

Bridge Dictionary

| Domain A | Bridge Theorem | Domain B |

|-----

The Grand Unification Chain

$\mathbb{R}^2$?Euclid??? Pythagorean Triples $\mathbb{Q}^2$ Light Cone

?

X. APPLICATIONS

| Application | Core Theorem | Status |

|-----

XI. OPEN PROBLEMS

| # | Problem | Status |

|---|-

Chapter 1.3

Source: *THEOREM_CATALOG.md*

The Pythagorean Cosmos: 3,158 Machine-Verified Theorems in Lean 4

Project: Pythagorean Cosmos Formal Mathematics Library

Table of Contents

1. Pythagorean Triples & the Berggren Tree

1. Pythagorean Triples & the Berggren Tree

Files: Basic.lean, Berggren.lean, BerggrenTree.lean, PythagoreanTriples.lean, PythagoreanPairing.lean,
PythagoreanNeuralArch.lean, AgentAlpha_Invariants.lean, AgentBeta_TreeDynamics.lean,
AgentEpsilon_Synthesis.lean, ParentDescent.lean, FermatFactor.lean

Core Parametrization

| Theorem | Statement | File |

|-----

Berggren Tree Structure

| Theorem | Statement | File |

|-----

Tree Dynamics & Invariants

| Theorem | Statement | File |

|-----

Pairing & Sum-of-Squares

| Theorem | Statement | File |

|-----

2. Four-Channel Integer Signatures

Files: ChannelEntropy.lean, PrimeSignatures.lean, Multiplicativity.lean, Defs.lean, Computations.lean, Session2Theorems.lean

Channel Formulas for Primes

| Theorem | Statement | File |

|-----

Signature Classes & Gap Theorem

| Theorem | Statement | File |

|-----

Multiplicative Structure

| Theorem | Statement | File |

|-----

Eisenstein Connection

| Theorem | Statement | File |

|-----

Divisibility

| Theorem | Statement | File |

|-----

Character Theory

| Theorem | Statement | File |

Powers of 2

| Theorem | Statement | File |

3. Compression Theory & Information Limits

Files: Compression.lean, CompressionTheory.lean, CompressionExtensions.lean, CodingTheory.lean, Entropy.lean

Fundamental Impossibility

| Theorem | Statement | File |

Extended Theory

| Theorem | Statement | File |

4. Quantum Computing & Gate Synthesis

Files: QuantumBerggren.lean, QuantumGateSynthesis.lean, QuantumCircuits.lean, QuantumGateAlgebra.lean, QuantumBerggrenResearch.lean, QuantumBerggrenGates.lean, QuantumCompression.lean, QuantumGates.lean, QuantumFoundations.lean, QuantumSimulation.lean, QuantumStructures.lean, QuantumTypeTheory.lean

Berggren Gate Group

| Theorem | Statement | File |

|-----

Theta Group Connection

| Theorem | Statement | File |

|-----

Circuit Theory

| Theorem | Statement | File |

|-----

Factoring via Gate Circuits

| Theorem | Statement | File |

|-----

5. Stereographic Projection & the Universal Decoder

Files: StereographicProjection.lean, StereographicDecoder.lean, StereographicRationals.lean,
InverseStereoMobius.lean, InverseStereoMobiusNext.lean, InverseStereoResearch.lean, RosettaStone.lean,
UniversalDecoder.lean, DimensionalProjection.lean, CMBLandscape.lean, SphericalCombination.lean

Core Properties

| Theorem | Statement | File |

|-----

Rational Arithmetic

| Theorem | Statement | File |

|-----

6. Fermat's Last Theorem (n=4) & Congruent Numbers

Files: FLT4.lean, CongruentNumber.lean

FLT4

| Theorem | Statement | File |

|-----

Congruent Numbers

| Theorem | Statement | File |

|-----

7. Lorentz Geometry & Light Cone Theory

Files: LightCone.lean, LightConeTheory.lean, PhotonParity.lean, PhotonResearchRound2.lean?Round5.lean,
PhotonFrontier.lean

Lorentz Structure

| Theorem | Statement | File |

|-----

Photon Statistics

| Theorem | Statement | File |

|-----

Non-Associativity

| Theorem | Statement | File |

|-----

8. Algebraic Structures & Cayley-Dickson Hierarchy

Files: CayleyDickson.lean, DivisionAlgebras.lean, BrahmaguptaFibonacci.lean, Algebra.lean, AlgebraicStructures.lean

Composition Identities

| Theorem | Statement | File |

|-----

Norm Closure

| Theorem | Statement | File |

|-----

Sum-of-Squares Filter

| Theorem | Statement | File |

|-----

9. $SL(2, ?)$ & Modular Group Theory

Files: SL2Theory.lean, Moonshine.lean, MillenniumConnections.lean, MillenniumDeep.lean

SL(2) Structure

| Theorem | Statement | File |

|-----

Moonshine

| Theorem | Statement | File |

|-----

10. Number Theory (Classical & Advanced)

Files: NumberTheory.lean, NumberTheoryAdvanced.lean, NumberTheoryDeep.lean, DeepResults.lean,
GaussianIntegers.lean, QuadraticForms.lean, Extensions.lean, NewTheorems.lean, ArithmeticGeometry.lean,
DiophantineApproximation.lean

Euler's Totient

| Theorem | Statement | File |

|-----

Möbius Function

| Theorem | Statement | File |

|-----

Cyclotomic Polynomials

| Theorem | Statement | File |

|-----

Primes

| Theorem | Statement | File |

|-----

Gaussian Integers

| Theorem | Statement | File |

|-----

Wilson's Theorem (Computational)

| Theorem | Statement | File |

|-----

11. Combinatorics & Graph Theory

Files: Combinatorics.lean, SauerShelah.lean, RamseyTheory.lean, ExtremalGraphTheory.lean, AdditiveCombinatorics.lean, SpectralGraphTheory.lean, GraphTheoryExploration.lean, MatroidTheory.lean

Ramsey Theory

| Theorem | Statement | File |

|-----

Euler's Polyhedra

| Theorem | Statement | File |

|-----

Graph Theory

| Theorem | Statement | File |

|-----

Sauer-Shelah (Open)

| Theorem | Statement | File |

|-----

12. Topology & Dynamical Systems

Files: Topology.lean, TopologyDynamics.lean, TopologyExploration.lean, DynamicalSystems.lean, ErgodicTheory.lean, AlgebraicTopology.lean, KnotTheory.lean, DifferentialGeometry.lean, SymplecticGeometry.lean, MetricGeometry.lean, ConvexGeometry.lean

Fixed Points

| Theorem | Statement | File |

|-----

Countability

| Theorem | Statement | File |

|-----

13. Category Theory & Representation Theory

Files: CategoryTheory.lean, CategoryTheoryDeep.lean, CategoryTheoryExploration.lean, CategoryRepresentation.lean, RepresentationTheory.lean, RepTheoryDeep.lean, HomologicalAlgebra.lean

Functorial

| Theorem | Statement | File |

|-----

14. Möbius Transformations & Order Classification

Files: OrderClassification.lean, IntegerChains.lean, Hypotheses.lean, InverseStereoMobius.lean

Order Classification (Novel Results)

| Theorem | Statement | File |

|-----

15. Crystallizer & Neural Architecture Theory

Files: CrystallizerMath.lean, CrystallizerFormalization.lean, CrystallizerFrontier.lean, NeuralCrystallizerFrontier.lean, PythagoreanNeuralArch.lean, InsideOutFactor.lean, InsideOutResearch.lean, EnergyDescentResearch.lean, IOFCore.lean, IOFDynamical.lean, IOFExplorations.lean

Energy Descent

| Theorem | Statement | File |

|-----

16. Applied Mathematics

Files: DriftFreeIMU.lean, HomingMissile.lean, ECDLP.lean, Applications.lean, RealWorldApplications.lean, CryptographyApplications.lean, CryptographyFoundations.lean, O1Impossibility.lean, NumericalAnalysis.lean

IMU Drift Correction

| Theorem | Statement | File |

|-----

Elliptic Curve Cryptography

| Theorem | Statement | File |

|-----

17. Advanced Topics

Files covering: Hodge theory, operator algebras, stochastic processes, information geometry, model theory, computability theory, tropical geometry, descriptive set theory, algebraic K-theory, geometric group theory

These files contain formalized foundations and key results in each area, totaling hundreds of additional theorems connecting to the core Pythagorean-quantum framework.

Statistics Summary

| Category | Files | Approx. Theorems |

|-----

Appendix: Duplicate Theorem Index

The following 118 theorem names appear in multiple files, representing independent proofs of the same statement in different mathematical contexts. The most duplicated theorem is brahmagupta_fibonacci (the Brahmagupta-Fibonacci two-square identity), appearing 16 times, reflecting its central role as the algebraic backbone connecting Pythagorean triples, Gaussian integers, photonic composition, and quantum gate synthesis.

Top duplicates by count:

*

These duplications are not errors but rather demonstrate how the same algebraic identities recur across the project's different mathematical domains ? a manifestation of the project's central thesis that a single algebraic thread (norm multiplicativity over the Cayley-Dickson hierarchy) unifies number theory, quantum computing, and mathematical physics.